

TOSHIBA TS3000 SERIES INSTRUCTION MANUAL



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Movej()

Moves()

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SetGlobalPYLOAD()

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SetPayload()

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Other ManualsLib Projects

TS3000 series Robot Controller

TS3000 SCARA / LINEAR system
TS3100 SCARA / LINEAR system
TSL3000 SCARA system

INSTRUCTION MANUAL TsRemoteLib.DLL MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of Toshiba Machine's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

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Preface

The TS3000/TSL3000 controller has the HOST function to enable the remote control of the robot by sending and receiving commands.

“TsRemoteLibS.DLL” compiles the programs that send to and receive commands from WindowsPC in a library to facilitate software development.

This manual explains how to use “TsRemoteLibS.DLL”.

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- Operator's Manual
- Robot Language Manual
- Interface Manual

This manual is structured as follows:

Section 1: Overview... Outlines the communication functions.

Section 2: Specification and Setting of the COM1 Port and the Host Port... Describes the hardware interface and communication mode of the RS-232C port.

Section 3: Specification and Setting of Ethernet... Describes the hardware interface and communication mode of Ethernet.

Section 4: Non-procedure Communication... Describes the procedure for the communication that only handles exchange between external devices and robot programs.

Section 5: Simple Procedure Communication ... Describes the procedure computer, commands and robot program files for the communication that targets a host.

Section 6: Robot Operation Sequence by External Communication ... Shows examples of operating the robot.

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Section 1 Overview

To control the robot in remote mode through communication with the TS3000/TSL3000 series robot controller, commands are sent to and received from the controller. “TsRemoteLib.DLL” compiles in DLL (Dynamic Link Library) the method groups for remote control of the robot in a simple manner without coding communication programs. “TsRemoteLib.DLL” runs under Windows .Net 2.0/3.0/3.5 and the applications are developed using VB/C++/C# of Microsoft Visual Studio.

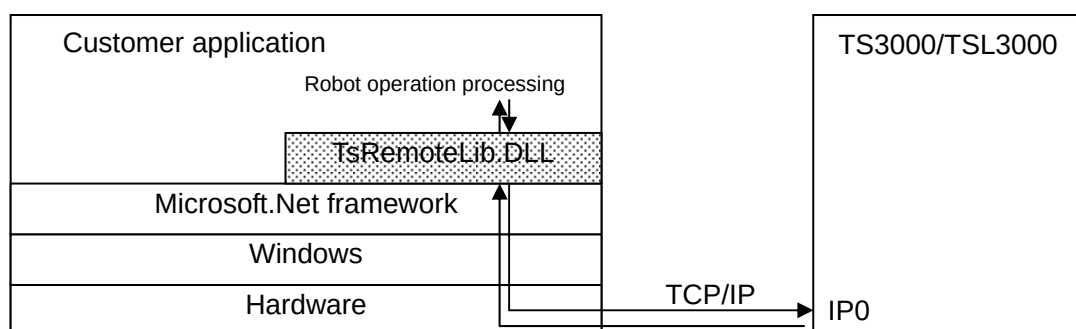


Fig. 1.1 TsRemoteLibS.DLL

The following diagram shows a coding example of a simple application for a robot operation command and stop on the buttons that are assigned in the form.

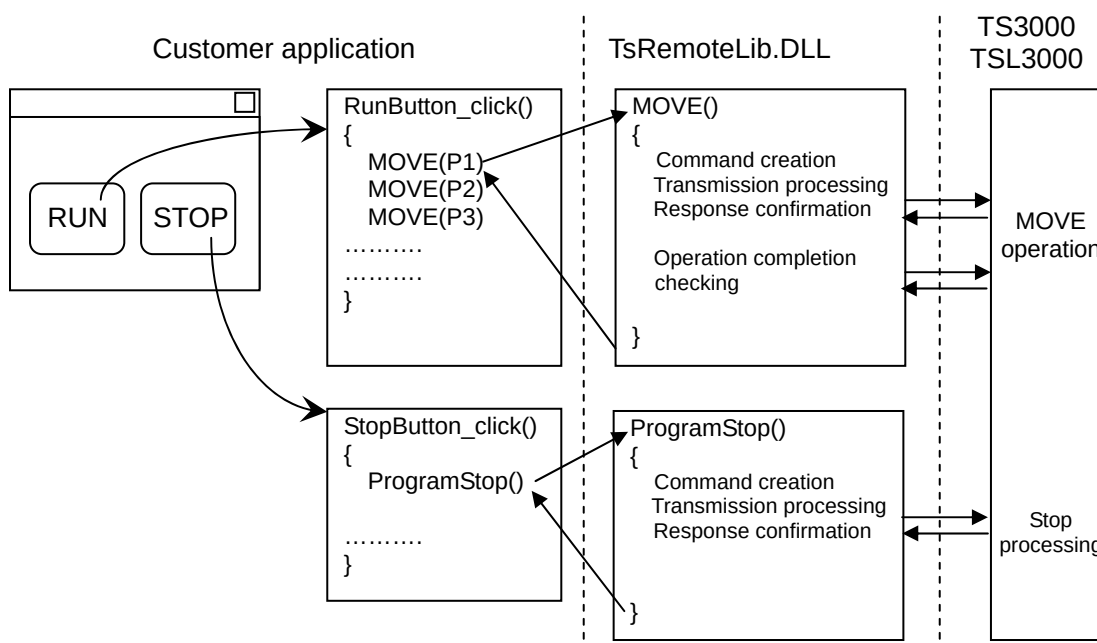


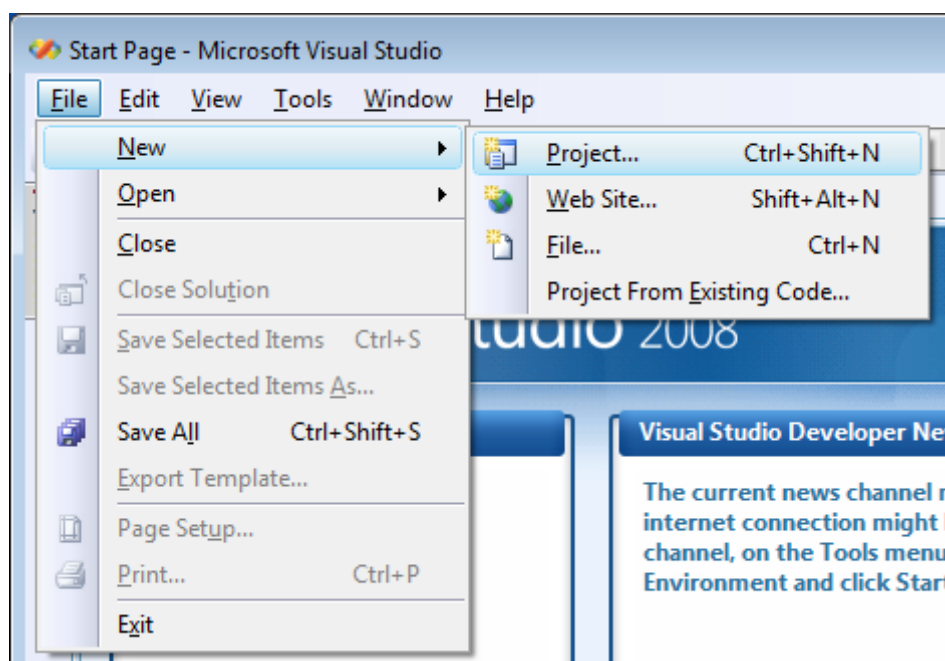
Figure 1.2 Outline of the application development using DLL

Section 2 Example of Application Development Using C#

This section describes the procedure for reference setting of TsRemoteLib for the application for controlling the SCARA robot. As the precondition, Visual Studio 2008 (referred to as VS2008 henceforth) must have been installed in the application creation environment. The file called TsRemoteLib.DLL is stored in the same folder as that of the application execution file.

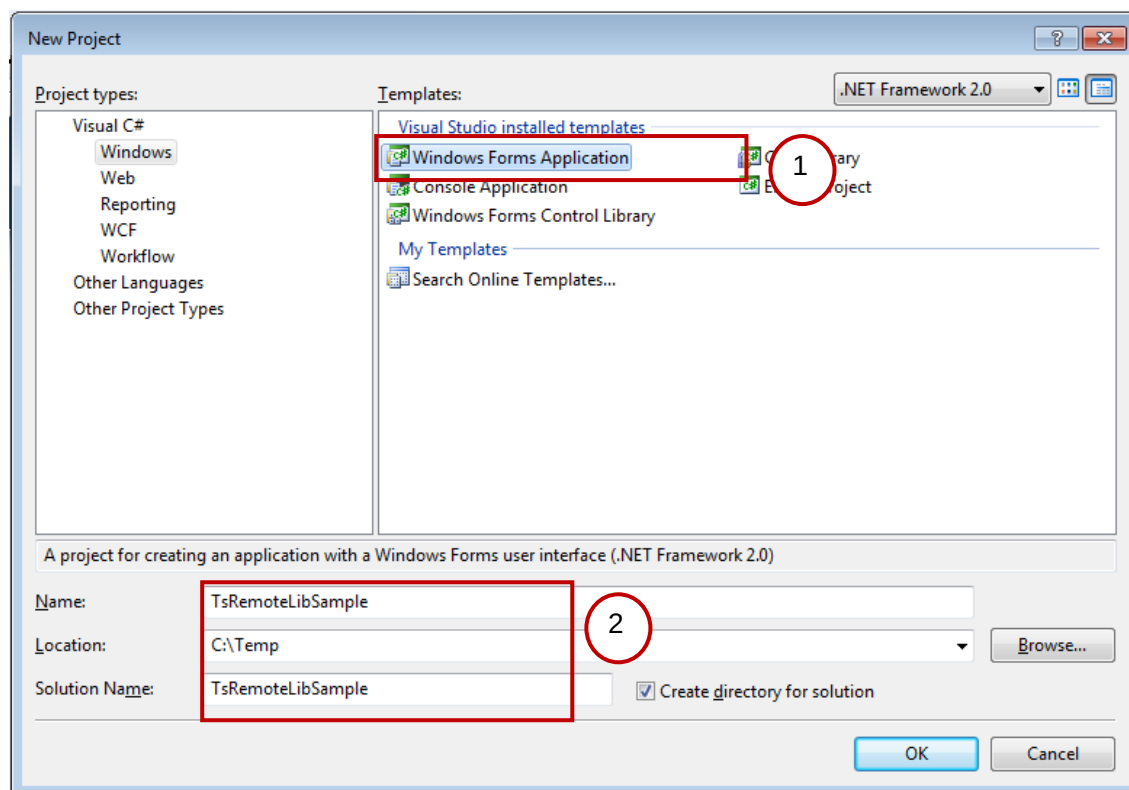
2.1 Creating an Application Solution

(1) Select File -> New -> Project.



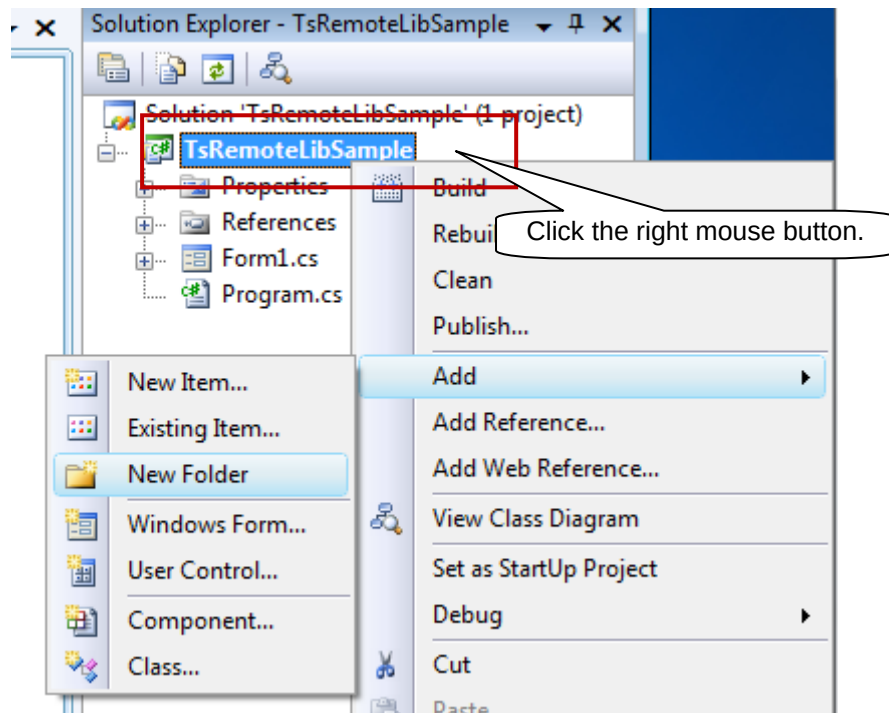
(2) Enter a project name.

- ① Select Windows Forms Application as the template.
- ② Enter a project name and the location.

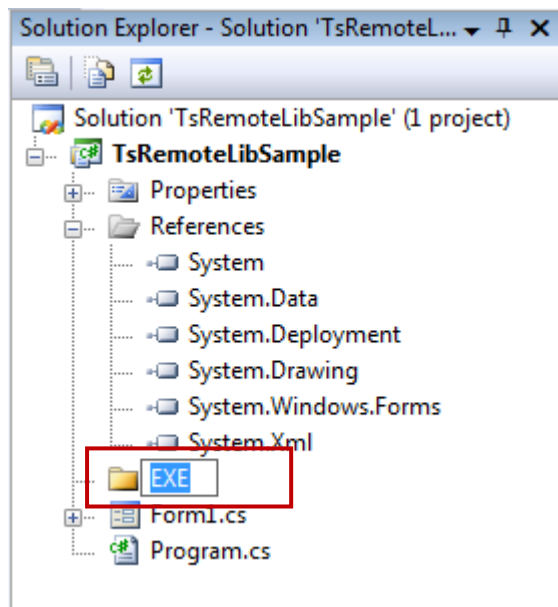


2.2 Setting Application Execution File Output Destination

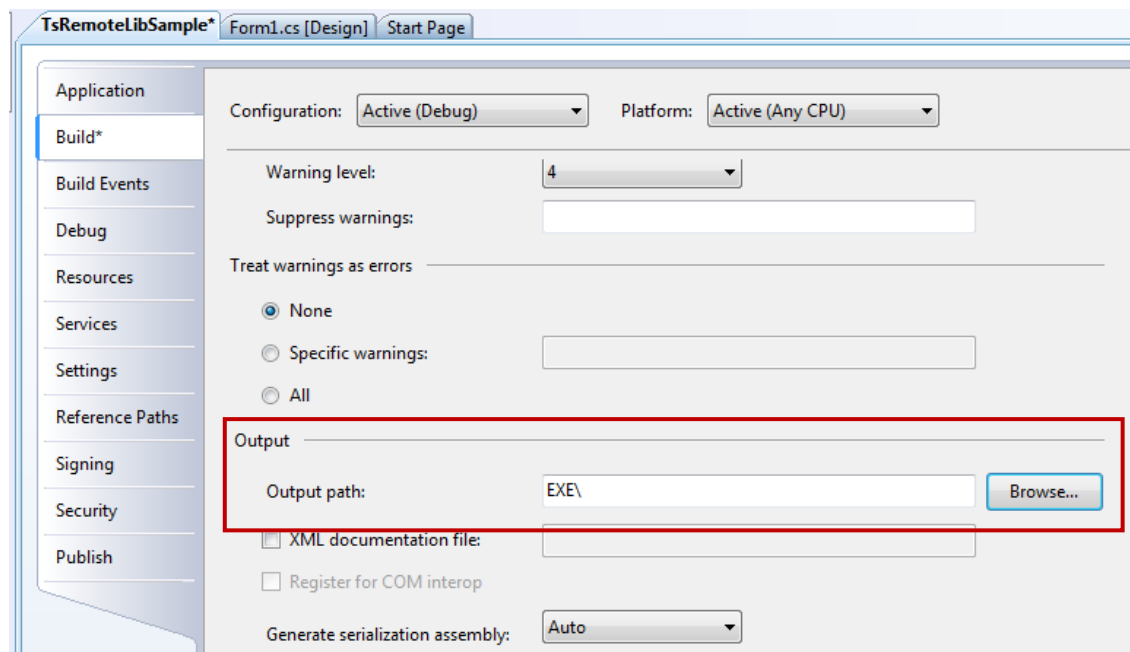
- (1) Set the mouse cursor on the required item with the Solution Explorer and click the right button.
- (2) Add the context menu that is displayed -> Select New folder.



- (3) Set EXE as the new folder name. (Assign any name.)



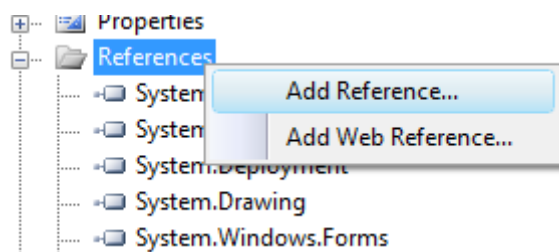
- (4) Set the execution file output destination to the EXE folder that was created on the Properties screen of the TsRemoteLibSample project.



- (5) Copy TsRemoteLib.DLL into the EXE folder that was created.
(TsRemoteLib.DLL and the application that is created by the customer (EXE file) must be stored in the same directory.)

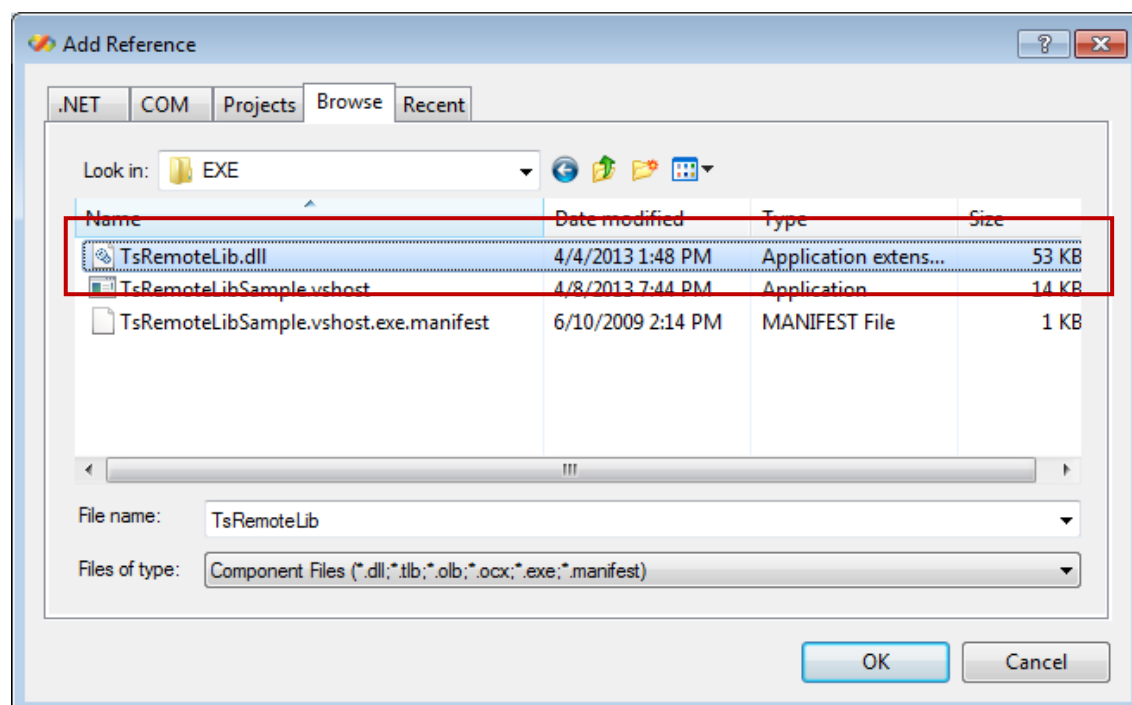
2.3 Reference Setting of TsRemoteLib

(1) Select Reference -> Add Reference.



(2) Select TsRemoteLib.DLL that is copied from the EXE folder and press OK.

TsRemoteLib can be used by setting reference.



2.4 Calling TsRemoteS Interface

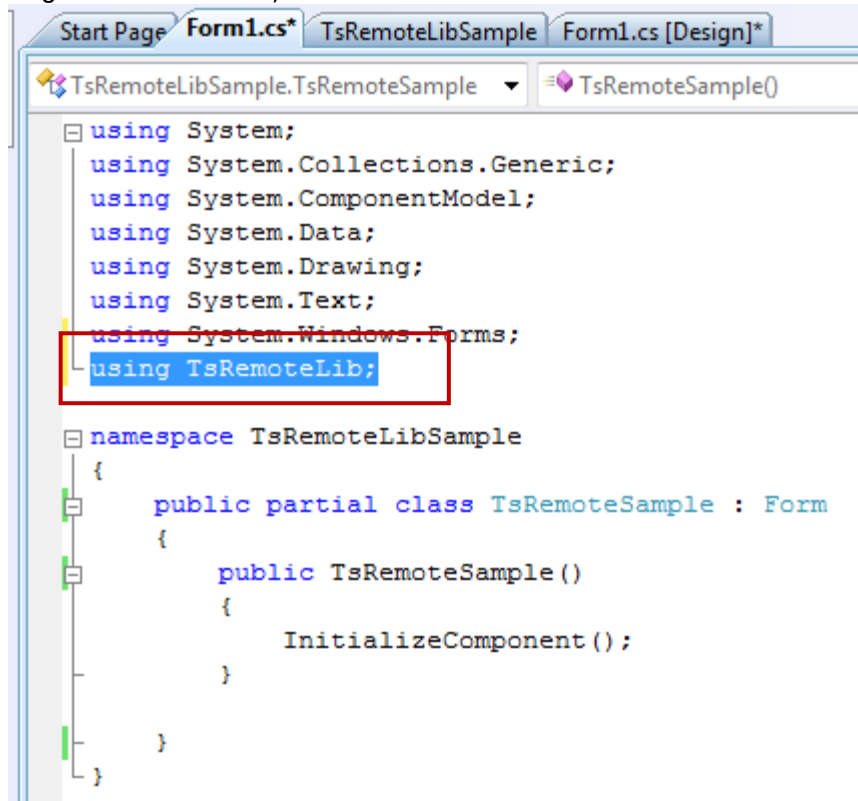
(1) Opening TsRemoteLibSample.cs

Form1 is used as the default name immediately after creation of a project. In this explanation, the form name is changed to **TsRemoteLibSample**.

(2) Declaring TsRemoteLib

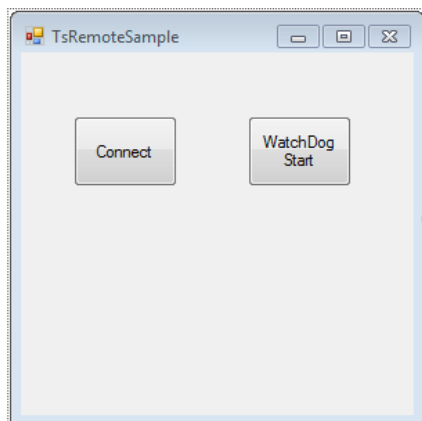
Enter the following command in TsRemoteLibSample.cs.

using TsRemoteLib;



(3) Calling the interface

Assume that the form contains the buttons that are shown below. The procedure for calling the interface is described in the source of each click event handler.



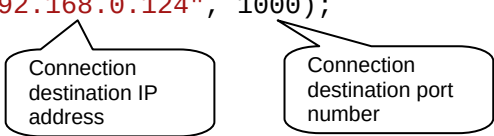
① Install connection button processing.

Use the instance name of class TsRemoteS for the Robot.

(In the case of 6-axis, Use the instance name of class TsRemoteV)

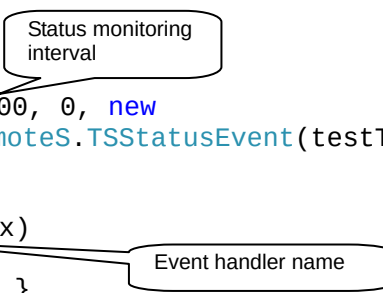
Sample IP address and port number are used.

```
private void btnConnect_Click(object sender, EventArgs e)
{
    try
    {
        if (Robot == null)
        {
            Robot = new TsRemoteS();
            Robot.SetIPAddr(0, "192.168.0.124", 1000);
        }
        Robot.Connect(1);
        //Connected successfully
    }
    catch (TsRemoteSException ex)
    {
        //Error processing
    }
}
```



② Start Watch Dog.

```
private void btnStartWatchDog_Click(object sender, EventArgs e)
{
    try
    {
        if (Robot != null)
        {
            Robot.WatchDogStart(100, 0, new
            TsRemoteS.TSStatusEvent(testTSStatusEvent));
        }
    }
    catch (TsRemoteSException ex)
    {
        //Error processing }
    }
}
```



2.5 Event Processing Performed Based on the Information from WatchDog of TsRemoteLib

Through WatchDog monitoring, information such as emergency stop, safety switch, mode, and Run status is acquired periodically. When the acquired information changes, an event occurs. Since the application basically runs in a different thread from that of WatchDog, create the handler for processing events from WatchDog as follows.

```
public void testTSStatusEvent(TsStatusMonitor para)
{
    if (InvokeRequired)
    {
        BeginInvoke( new TsRemoteS.TSStatusEvent(testTSStatusEvent), para);
    }
    else
    {
        //Evaluate the value of TsStatusMonitor and perform event processing.
        if (para.EmergencyStop == 1)
        {
            //Processing when the Emergency Stop button is pressed.
            . . . . .
        }
        if (para.SafetyStop == 1)
        {
            //Processing when the safety switch is pressed.
            . . . . .
        }
        . . . . .
        . . . . .
    }
}
```

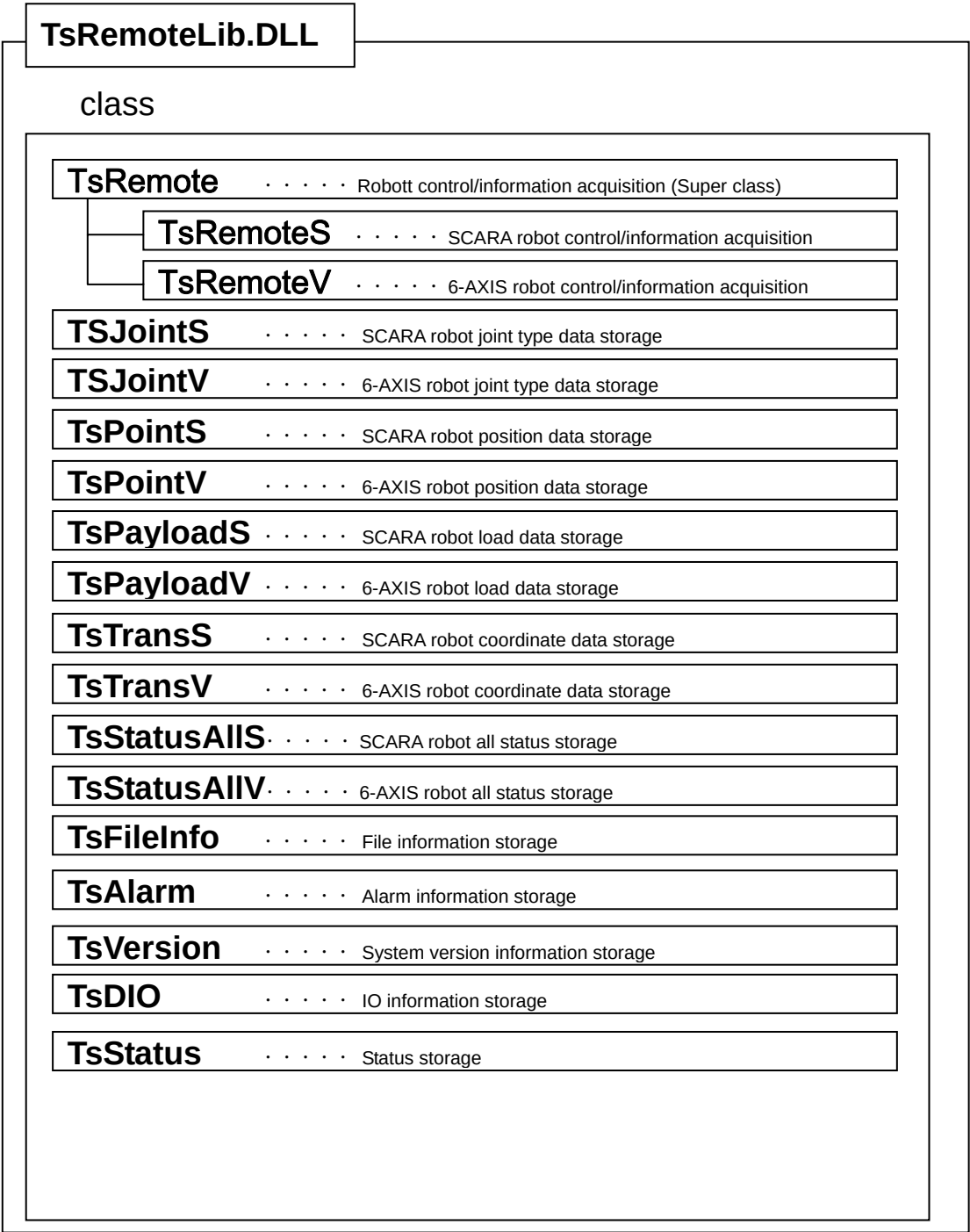
Event occurrence factors handled by WatchDog are listed below. Refer to the TsStatusMonitor class for the details.

- Emergency Stop request
- Safety switch stop request
- Various mode statuses
- Override
- Alarm
- Servo status
- WatchDog error
- Ethernet communication error

Section 3 Reference

3.1 Outline of TsRemoteLib

The following classes are installed in TSRemoteLib for remote control of the robot and information acquisition.



3.2 Classes That Are Defined in TsRemoteLib

TsRemote

class

Classes for remote-controlling the SCARA and 6-AXIS robot

TsRemote class is super class of the TsRemoteS/TsRemoteV class
 In the case of SCARA robot, make the instance of the TsRemoteS class.
 In the case of 6-AXIS robot, make the instance of the TsRemoteV class.

TsRemoteS

class

Classes for remote-controlling the SCARA and 6-AXIS robot

In the case of SCARA robot, make the instance of the TsRemoteS class.
 Please refer to Clause "3.4 TsRemoteS Class Properties".
 Please refer to Clause "3.5 TsRemoteS Class Method".

TsRemoteV

class

Classes for remote-controlling the 6-AXIS robot

In the case of 6-AXIS robot, make the instance of the TsRemoteV class.
 Please refer to Clause "3.4 TsRemoteV Class Properties".
 Please refer to Clause "3.5 TsRemoteV Class Method".

TsPointS

class

Class for storing Point data of SCARA robot

Property

double X	X point/speed
double Y	Y point/speed
double Z	Z point/speed
double C	C point/speed
double T	T point/speed
enum ConfigS	Configuration (posture)

See also

GetGlobalPoint(), GetPsnCmdWorld(), SetPsnCmdWorld(), Move(), Moves(), Movej()

TsPointV

class

Class for storing Point data of 6-AXIS robot

Property

double	X	X point/speed
double	Y	Y point/speed
double	Z	Z point/speed
double	A	A point/speed
double	B	B point/speed
double	C	C point/speed
double	U	U point/speed
double	V	V point/speed
ConfigVpart	Config	Configuration (posture)

See also

GetGlobalPoint(),GetPsnCmdWorld(),SetPsnCmdWorld(),Move(),Moves(),Movej()

ConfigVpart

class

Class for for storing configuration (posture) of 6-AXIS robot

Property

TvSholder	Sholder	Sholder configuration(enum 0:Free / 1:Lefty / 2:Righty)
TvElbow	Elbow	Elbow configuration(enum 0:Free / 1:Above / 2:Below)
TvWrist	Wrist	Wrist configuration(enum 0:Free / 1:Nflip / 2:Flip)
TvRot4	Rot4	4th axis configuration(enum 0:Free / 1:Single4 / 2:Double4)
TvRot6	Rot6	6th axis configuration(enum 0:Free / 1:Single6 / 2:Double6)
ConfigV	All	(enum configV)

About Config

config of the 6-Axis robot is decided by the combination of five elements of Sholder, Elbow, Wrist, Rot4 and Rot6.

"All property "can set five elements by a lump. (enum ConfigV)

See also

TsPointV

TsJointS

class

Class for storing Joint data of SCARA robot

Property

double J1	1 st joint position
double J 2	2 nd joint position
double J3	3 rd joint position
double J4	4 th joint position
double J5	5 th joint position

See also

TsStatusAllS, GetPsnCmdJoint(), GetPsnFbkJoint(), GetVelocJoint()

TsJointV

class

Class for storing Joint data of 6-AXIS robot

Property

double J1	1 st joint position
double J 2	2 nd joint position
double J3	3 rd joint position
double J4	4 th joint position
double J5	5 th joint position
double J6	6 th joint position
double J7	7 th joint position
double J8	8 th joint position

See also

TsStatusAllV, GetPsnCmdJoint(), GetPsnFbkJoint(), GetVelocJoint()

TsTransS

class

Class for storing coordinate data of SCARA robot

Property

double X	X coordinate
double Y	Y coordinate
double Z	Z coordinate
double C	C coordinate

See also

GetGlobalTRANS(), SetGlobalTRANS(), SetTransTool(), SetTransWork()

TsTransV

class

Class for storing coordinate data of 6-AXIS robot

Property

double X	X coordinate
double Y	Y coordinate
double Z	Z coordinate
double A	A coordinate
double B	B coordinate
double C	C coordinate

See also

GetGlobalTRANS(), SetGlobalTRANS(), SetTransTool(), SetTransWork()

TSPayloadS

class

Class for storing point data of SCARA robot

Property

double M	Weight (kg)
double L	Offset of the center of gravity (mm)

See also

SetGlobalPYLOAD(), GetGlobalPAYLOAD, SetPYLOAD()

TSPayloadV

class

Class for storing point data of 6-AXIS robot

Property

double M	Weight (kg)
double Lx	Offset of the center of gravity (mm)
double Ly	Offset of the center of gravity (mm)
double Lz	Offset of the center of gravity (mm)
double Gx	Work size(mm)
double Gy	Work size(mm)
double Gz	Work size(mm)

See also

SetGlobalPYLOAD(), GetGlobalPAYLOAD, SetPYLOAD()

TsFileInfo

class

Class for storing file information that is stored in the controller

Property

string name	File name
int size	Size

See also

GetDir()

TsVersion

class

Class for storing a controller system version

Property

string SystemName	System name
DateTime Date	System creation date and time
string Sum	Check Sum (maximum number of characters)

See also

GetSystemVersion()

TsDIO

class

Class for storing input/output status of the controller

*Property***Int16 Din[20]**

Input signal status

Handling bits of Din data and input signals

Din[0] : DIN1 ~ DIN16 Standard input
 Din[1] : DIN17 ~ DIN32 Standard input
 Din[2] : DIN33 ~ DIN48 Standard input
 Din[3] : DIN49 ~ DIN64 Standard input
 Din[4] : DIN101 ~ DIN116 Extended input
 Din[5] : DIN117 ~ DIN132 Extended input
 Din[6] : DIN133 ~ DIN148 Extended input
 Din[7] : DIN149 ~ DIN164 Extended input
 Din[8] : DIN201 ~ DIN216 System input
 Din[9] : DIN217 ~ DIN232 System input
 Din[10] : DIN233 ~ DIN248 System input
 Din[11] : DIN249 ~ DIN264 System input
 Din[12] : DIN301 ~ DIN316 Field bus input
 Din[13] : DIN317 ~ DIN332 Field bus input
 Din[14] : DIN333 ~ DIN348 Field bus input
 Din[15] : DIN349 ~ DIN364 Field bus input
 Din[16] : DIN401 ~ DIN416 Field bus input
 Din[17] : DIN417 ~ DIN432 Field bus input
 Din[18] : DIN433 ~ DIN448 Field bus input
 Din[19] : DIN449 ~ DIN464 Field bus input

Int16 Dout[20]

Output signal status

Handling bits of Dout data and output signals

Dout[0] : DOUT1 ~ DOUT16 Standard output
 Dout[1] : DOUT17 ~ DOUT32 Standard output
 Dout[2] : DOUT33 ~ DOUT48 Standard output
 Dout[3] : DOUT49 ~ DOUT64 Standard output
 Dout[4] : DOUT101 ~ DOUT116 Extended output
 Dout[5] : DOUT117 ~ DOUT132 Extended output
 Dout[6] : DOUT133 ~ DOUT148 Extended output
 Dout[7] : DOUT149 ~ DOUT164 Extended output
 Dout[8] : DOUT201 ~ DOUT216 System output
 Dout[9] : DOUT217 ~ DOUT232 System output
 Dout[10] : DOUT233 ~ DOUT248 System output
 Dout[11] : DOUT249 ~ DOUT264 System output
 Dout[12] : DOUT301 ~ DOUT316 Field bus output
 Dout[13] : DOUT317 ~ DOUT332 Field bus output
 Dout[14] : DOUT333 ~ DOUT348 Field bus output
 Dout[15] : DOUT349 ~ DOUT364 Field bus output
 Dout[16] : DOUT401 ~ DOUT416 Field bus output
 Dout[17] : DOUT417 ~ DOUT432 Field bus output
 Dout[18] : DOUT433 ~ DOUT448 Field bus output
 Dout[19] : DOUT449 ~ DOUT464 Field bus output

See also

GetDIO()

TsStatus

class

Class for storing Robot statuses

Property

int MasterMode	Master mode status 0: Teaching 1: Internal 2: Ext.Sig 3: Ext.Host 4: Ext.232C 5: Ext.Ether
int RunMode	Run mode status 0: CONT 1: CYCLE 2: STEP 3: SEGMENT
string SelectFile	Name of the program that is currently selected by the controller
int ovrd	Current override value Value range: 1%~100%
int MachineLock	Reserved (currently not used) Always set to 1
int RunStatus	Run status 0: STOP(RESET) 1: RUN 2: STOP(RETRY) 3: STOP(CONT)

See also

GetStatus()

TsStatusAllS

class

Class for storing all the statuses of SCARA robot

Property

int ServoStatus	Servo status 0:OFF 1:ON
int EmergencySwitch	Emergency Stop switch status 0:OFF 1:ON
int RunStatus	Run status 0:STOP(RESET) 1:RUN 2:STOP(RETRY) 3:STOP(CONT)
int SUCommandReq	Command request 0:None 1:Request occurred
string AlarmNo[10]	Current alarm (max. 10) Stored in a character string format such as "8-014". When there is no alarm, all the 10 items are set to null.
int ProgramExeLine	Program line number that is currently being executed
int ProgramAnalyzeLine	Program line number that is currently being analyzed
int ProgramExeTaskNo	Task number that is currently being executed
int ProgramAnalyzeTask	Task number that is currently being analyzed
int FeedHold	Feed Hold status 0:OFF 1:ON
int JogCoordinate	JOG-guided coordinate selection status 0:Joint 1:Tool 2:Work 3:World
int JogSpeed	JOG-guided guiding speed setting status 0:Low 1:Medium 2:High
int JogRemote	JOG-guided guiding mode setting status 0:Jog 1:Inching 2:Free
int MasterMode	Master mode status 0:Teaching 1:Internal 2:Ext. Sig 3:Ext. Host(Ext.232C/ExtEther)

TsStatusAllS property(continued)

int PowerOnTime Power On time (min)

int ProgramExecTime Program execution time (min)

Int16 Din[16] Input signal status
 Handling bits of Din data and input signals
 Din[0] : DIN1 ~ DIN16 Standard input
 Din[1] : DIN17 ~ DIN32 Standard input
 Din[2] : DIN33 ~ DIN48 Standard input
 Din[3] : DIN49 ~ DIN64 Standard input
 Din[4] : DIN101 ~ DIN116 Extended input
 Din[5] : DIN117 ~ DIN132 Extended input
 Din[6] : DIN133 ~ DIN148 Extended input
 Din[7] : DIN149 ~ DIN164 Extended input
 Din[8] : DIN201 ~ DIN216 System input
 Din[9] : DIN217 ~ DIN232 System input
 Din[10] : DIN233 ~ DIN248 System input
 Din[11] : DIN249 ~ DIN264 System input
 Din[12] : DIN301 ~ DIN316 Field bus input
 Din[13] : DIN317 ~ DIN332 Field bus input
 Din[14] : DIN333 ~ DIN348 Field bus input
 Din[15] : DIN349 ~ DIN364 Field bus input

Int16 Dout[16] Output signal status
 Handling bits of Dout data and output signals
 Dout[0] : DOUT1 ~ DOUT16 Standard output
 Dout[1] : DOUT17 ~ DOUT32 Standard output
 Dout[2] : DOUT33 ~ DOUT48 Standard output
 Dout[3] : DOUT49 ~ DOUT64 Standard output
 Dout[4] : DOUT101 ~ DOUT116 Extended output
 Dout[5] : DOUT117 ~ DOUT132 Extended output
 Dout[6] : DOUT133 ~ DOUT148 Extended output
 Dout[7] : DOUT149 ~ DOUT164 Extended output
 Dout[8] : DOUT201 ~ DOUT216 System output
 Dout[9] : DOUT217 ~ DOUT232 System output
 Dout[10] : DOUT233 ~ DOUT248 System output
 Dout[11] : DOUT249 ~ DOUT264 System output
 Dout[12] : DOUT301 ~ DOUT316 Field bus output
 Dout[13] : DOUT317 ~ DOUT332 Field bus output
 Dout[14] : DOUT333 ~ DOUT348 Field bus output
 Dout[15] : DOUT349 ~ DOUT364 Field bus output

TsJointS JointPosition SCARA Robot current joint position

TsPointS WorldPosition Current position of the SCARA robot world coordinate system

TsPointS WorkPosition Current position of the SCARA robot work coordinate system

string WorkName Name of the work coordinate system that is currently selected

string ToolName Name of the tool coordinate system that is currently selected

string BaseName Name of the base coordinate system that is currently selected

See also

GetStatusAll()

TsStatusAllV

class

Class for storing all the statuses of 6-AXIS robot

Property

int ServoStatus	Servo status 0:OFF 1:ON
int EmergencySwitch	Emergency Stop switch status 0:OFF 1:ON
int RunStatus	Run status 0:STOP(RESET) 1:RUN 2:STOP(RETRY) 3:STOP(CONT)
int SUCommandReq	Command request 0:None 1:Request occurred
string AlarmNo[10]	Current alarm (max. 10) Stored in a character string format such as "8-014". When there is no alarm, all the 10 items are set to null.
int ProgramExeLine	Program line number that is currently being executed
int ProgramAnalyzeLine	Program line number that is currently being analyzed
int ProgramExeTaskNo	Task number that is currently being executed
int ProgramAnalyzeTask	Task number that is currently being analyzed
int FeedHold	Feed Hold status 0:OFF 1:ON
int JogCoordinate	JOG-guided coordinate selection status 0:Joint 1:Tool 2:Work 3:World
int JogSpeed	JOG-guided guiding speed setting status 0:Low 1:Medium 2:High
int JogRemote	JOG-guided guiding mode setting status 0:Jog 1:Inching 2:Free
int MasterMode	Master mode status 0:Teaching 1:Internal 2:Ext. Sig 3:Ext. Host(Ext.232C/ExtEther)

TsStatusAllS property(continued)

int PowerOnTime Power On time (min)

int ProgramExecTime Program execution time (min)

Int16 Din[16] Input signal status
 Handling bits of Din data and input signals
 Din[0] : DIN1 ~ DIN16 Standard input
 Din[1] : DIN17 ~ DIN32 Standard input
 Din[2] : DIN33 ~ DIN48 Standard input
 Din[3] : DIN49 ~ DIN64 Standard input
 Din[4] : DIN101 ~ DIN116 Extended input
 Din[5] : DIN117 ~ DIN132 Extended input
 Din[6] : DIN133 ~ DIN148 Extended input
 Din[7] : DIN149 ~ DIN164 Extended input
 Din[8] : DIN201 ~ DIN216 System input
 Din[9] : DIN217 ~ DIN232 System input
 Din[10]: DIN233 ~ DIN248 System input
 Din[11]: DIN249 ~ DIN264 System input
 Din[12]: DIN301 ~ DIN316 Field bus input
 Din[13]: DIN317 ~ DIN332 Field bus input
 Din[14]: DIN333 ~ DIN348 Field bus input
 Din[15]: DIN349 ~ DIN364 Field bus input

Int16 Dout[16] Output signal status
 Handling bits of Dout data and output signals
 Dout[0] : DOUT1 ~ DOUT16 Standard output
 Dout[1] : DOUT17 ~ DOUT32 Standard output
 Dout[2] : DOUT33 ~ DOUT48 Standard output
 Dout[3] : DOUT49 ~ DOUT64 Standard output
 Dout[4] : DOUT101 ~ DOUT116 Extended output
 Dout[5] : DOUT117 ~ DOUT132 Extended output
 Dout[6] : DOUT133 ~ DOUT148 Extended output
 Dout[7] : DOUT149 ~ DOUT164 Extended output
 Dout[8] : DOUT201 ~ DOUT216 System output
 Dout[9] : DOUT217 ~ DOUT232 System output
 Dout[10]: DOUT233 ~ DOUT248 System output
 Dout[11]: DOUT249 ~ DOUT264 System output
 Dout[12]: DOUT301 ~ DOUT316 Field bus output
 Dout[13]: DOUT317 ~ DOUT332 Field bus output
 Dout[14]: DOUT333 ~ DOUT348 Field bus output
 Dout[15]: DOUT349 ~ DOUT364 Field bus output

TsJointV JointPosition 6-AXIS Robot current joint position

TsPointV WorldPosition Current position of the 6-AXIS robot world coordinate system

TsPointV WorkPosition Current position of the 6-AXIS robot work coordinate system

string WorkName Name of the work coordinate system that is currently selected

string ToolName Name of the tool coordinate system that is currently selected

string BaseName Name of the base coordinate system that is currently selected

See also

GetStatusAll()

TsStatusMonitor

class

Class for storing the cause of the event that occurs under WatchDog status monitoring

*Property***int AlarmLevel**

Indicates the alarm level that is currently occurring. When the alarm level changes, an event occurs.

- 0: No alarm
- 1: Level 1 alarm occurred.
- 2: Level 2 alarm occurred.
- 4: Level 4 alarm occurred.
- 8: Level 8 alarm occurred.

int BreakCommand

Indicates that a BREAK request occurred to the controller during the period from the previous WatchDog monitoring request to the current monitoring request.

- 0:None
- 1:BREAK request occurred.

int EmergencyStop

Indicates that an Emergency Stop request occurred to the controller during the period from the previous WatchDog monitoring request to the current monitoring request. When this flag is set to 1, an event occurs.

- 0: None
- 1: Emergency Stop request occurred.

int EmergencySwitch

Indicates the current Emergency Stop status. When the status changes, an event occurs. Since WatchDog monitoring is performed at a specified interval, check the EmergencyStop property to capture the Emergency Stop request without fail.

- 0:OFF (The circuit is closed.)
- 1:ON (The circuit is opened.)

int EtherNetConnection

Indicates the Ethernet connection status. When an error occurs due to disconnection or so on, an event occurs.

- 0: No error
- 1: An error occurred in the Ethernet communication processing of Windows.

int MasterMode

Indicates the master mode status. When the status changes, an event occurs.

- 0:Teaching
- 1:Internal
- 2:Ext.Sig
- 3:Reservation
- 4:Ext232C
- 5:ExtEther

int Override

Indicates the override status (1-100%). When the value changes, an event occurs.

int RunMode	Indicates the robot run mode. When the mode changes, an event occurs. 0:CONT 1:CYCLE 2:STEP 3:SEGMENT
int RunStatus	Indicates the robot run status. When the status changes, an event occurs. 0:STOP(RESET) 1:RUN 2:STOP(RETRY) 3:STOP(CONT)
int SafetyStop	Indicates that a stop request occurred to the controller due to the Safety switch ON state during the period from the previous WatchDog monitoring request to the current monitoring request. 0: None 1: A stop request occurred to the controller due to the Safety switch ON state.
int SafetySwitch	Indicates the status of the Safety switch. Since WatchDog monitoring is performed at the specified interval, check the SafetyStop property to capture the Stop request by the Safety switch without fail. 0: OFF (The circuit is closed.) 1: ON (The circuit is open.)
int ServoStatus	Indicates the current Servo status. 0:Servo OFF 1:Servo ON
int StopCommand	Indicates that a Stop request occurred to the controller during the period from the previous WatchDog monitoring request to the current monitoring request. When this flag is set to 1, an event occurs. 0: None 1: STOP request occurred.
int WatchDogErr	Indicates the occurrence of a timeout error by WatchDog monitoring. 0: None 1: Timeout error occurred.

See also
GetStatusAll()

TsRemoteSException class

Class for deriving the Application Exception class that occurs at execution of each method of the TsRemoteS and TsRemoteV class

Property

ErrorCode errorCode Stores the error code that was handled. Refer to the ErrorCode enumeration for the details of the error code.

See also

ErrorCode

3.3 Enumerations That Are Defined in TsRemoteLib

ConfigS

enum

Enumeration for defining the SCARA robot configuration (posture)

Format
enum ConfigS

Value

FREE	0: Indicates that the robot configuration (posture) is FREE.
LEFTY	1: Indicates that the robot configuration (posture) is LEFTY.
RIGHTY	2: Indicates that the robot configuration (posture) is RIGHTY.

See also
MvConfig, Move(), Moves(), Movej(), Movea()

ConfigV

enum

Enumeration for defining the 6-Axis robot configuration (posture)

Format
enum ConfigV

Value
ConfigV can be set by string.(Ex. "xxxxx", "LANSS", "RBxSx")

Sholder	Elbow	Wrist	Rot4	Rot6
x:Free(0)	x:Free(0)	x:Free(0)	x:Free(0)	x:Free(0)
L:Lefty(1)	A:Above(1)	N:Nflip(1)	S:Single4(1)	S:Single6(1)
R:Righty(2)	B:Below(2)	F:Flip(2)	D:Double4(2)	D:Double6(2)

See also
MvConfig, Move(), Moves(), Movej(), Movea()

Gain

enum

Enumeration for defining of robot motor status "GAIN"

Format

enum Gain

Value

OFF	0:off
ON	1:on

See also

MvConfig, Move(), Movea(), Moves(), Movej()

RunMode

enum

Enumeration for defining the robot run status

Format

enum RunMode

Value

CONT	0: Indicates that the robot operation mode is CONT.
CYCLE	1: Indicates that the robot operation mode is CYCLE.
STEP	2: Indicates that the robot operation mode is STEP.
SEGMENT	3: Indicates that the robot operation mode is SEGMENT.

See also

ErrorCode

enum

Enumeration for defining the exception occurrence factor by exceptions

Format

enum ErrorCode

Value

ControllerNG	1: The controller returned "NG".
CommunicationError	2: Communication error occurred.
NoCommand	3: The command parameter is invalid.
NoSerialportData	4: Data has not reached the serial port.
NoIP_PORTData	5: Data has not reached the Ethernet port.
TCPConnectTimeout	6: Timeout occurred in the Connect() method.
TCPError	7: Exception occurred in the internal processing of the Connect() method.
TCPDisconnect	8: The Ethernet port was disconnected.
TCPNotConnect	9: The Ethernet port is not connected.
TCPReadError	10: Exception occurred in the DDL internal NetworkStream.Read method.
TCPWriteError	11: Exception occurred in the DDL internal NetworkStream.Write method.
TCPNoData	12: Data could not be received.
TCPReceiveTimeout	13: Timeout occurred in the transmission/reception processing of TCP communication.
IllegalIPAddressss	14: IP address setting error
IllegalStopbit	15: Stop bit setting error
IllegalDatabits	16: Data length setting error
IllegalCommType	17: Connection destination setting error
IllegalParam	18: The parameter specified in the argument is incorrect.
IllegalBps	19: Baud rate setting error
IllegalPortNo	20: Port number setting error
IllegalWatchDogInterval	21: WatchDog setting error
IllegalAlarmLevel	22: Alarm level setting error
CannotStartWatchdog	23: Move*() or DirectDo() was executed without WatchDog monitoring.
NotfoundDirectory	24: The specified directory does not exist in the PC.
NotFondFile	25: The specified file does not exist.
MoveStop	26: Move*() stopped in response to the STOP signal. The Move*() method was executed while internal flag MoveStop is ON.
MoveBreak	27: Move*() stopped in response to the BREAK signal.
MoveTimeout	28: Move*() was terminated due to timeout.
CannotDoMove	29: Move*() or DirectDo() was executed without WatchDog monitoring.

See also

3.4 TsRemoteS Class Properties

TsRemote class is base class of the TsRemoteS and TsRemoteV class.
The property of TsRemote class is succeeded to by TSRemoteS class /TSRemoteV class.

MvAccel

TsRemote.property

Specifies/acquires the acceleration speed of the operation that is instructed by Move(), Moves(), Movej(), and Movea().

Format

MvAccel

Value

double 1.0~100000.0%(initial value:100)

Notice

If MvAccel is set to a value exceeding 100%, a motor overload alarm may occur for the operation that involves significant movement distance.

See also

MvDecel, MvSpeed, MvConfig, Move(), Moves(), Movej(), Movea()

MvDecel

TsRemote.property

Specifies/acquires the deceleration speed of the operation that is instructed by Move(), Moves(), Movej(), and Movea().

Format

MvDecel

Value

double 1.0~100000.0%(initial value: 100.0)

Notice

If MvDecel is set to a value exceeding 100%, a motor overload alarm may occur for the operation that involves significant movements.

See also

MvAccel, MvSpeed, MvConfig, Move(), Moves(), Movej(), Movea()

MvSpeed

TsRemote.property

Specifies/acquires the speed of the operation that is instructed by Move(), Moves(), Movej(), and Movea().

Format

MvSpeed

Value

double 1.0~100.0%(initial value: 100)

Notice

See also

MvAccel, MvSpeed, MvConfig, Move(), Moves(), Movej(), Movea()

3.5 TsRemote Class Method

TsRemote class is base class of the TsRemoteS and TsRemoteV class.
The method of TsRemote class is succeeded to by TSRemoteS class /TSRemoteV class.

Connect()

TsRemote.method

Connects to the controller.

*Format***bool Connect(int type)***Argument***int type**

Specify a connection type.

0:RS-232C(function not yet installed)**1:**TCP/IP*Return value***bool****true:**Connect successful**false:**Connect failed*Exceptions*

TCPConnectTimeout

Timeout occurred in the Connect() method.

NoIP_PORTData

The IP address or port for TCP/IP has not been set

IllegalCommType

Illegal connection type.Should be 0 for SerialPort,or 1 for TCP/IP

NoSerialportData

The datas for serial communication have not been set

TCPError

The error has occurred in the network

CommunicationError

Communication Error

ControllerNG

Command not accepted by controller

IllegalParam

Illegal parameter

Notice

Connection under RS-232C(COM) is not supported.

See also

Disconnect()

DeleteFile()

TsRemote.method

Deletes the file that is stored in the robot controller.

*Format***void DeleteFile(string filename)***Argument***string** filename File name to be deleted from the robot controller*Return value*

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	

Notice

DeleteFile deletes the file in the controller unconditionally. When confirmation is necessary, display a conformation message by using an application.

See also

GetDir()

DirectDo()

TsRemote.method

Executes one line of program by using the direct execution function.

Format

void DirectDo(string command)

Argument

string command Program to be executed

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
CannotDoMove	Move command can not be executed, because of watch dog not running

Notice

DirectDo() cannot be executed unless monitoring is executed by the WatchDogStart() method. Execution of DirectDo() is completed at the commencement of the execution of the program after the controller receives the command. Completion of program operation cannot be checked. Use the Move*() method to execute an operation command such as MOVE.

When an event occurs due to the execution of EmergencyStop, SafetyStop, StopCommand, or BreakCommand, the DLL internal flag (MoveStop) is set to ON.

If the DirectDo() method is executed while the internal flag (MoveStop) is ON, MoveStop exception occurs, thereby restraining the robot operation. ResetMove() resets the internal flag, enabling the robot operation. Reset the operation after checking with the user through a dialog and so on.

See also

WatchDogStart(), ResetMove()

Disconnect()

TsRemote.method

Disconnects communication with the controller.

*Format***bool Disconnect()***Argument*

None

Return value

None

Exceptions

None

Notice

To terminate a customer-created application, terminate application after disconnecting the communication with the controller by executing Disconnect().

See also

Connect()

Dispose()

TsRemote.method

Disposes of the TsRemoteS object.

Format

void Dispose()

Argument

None

Return value

None

Exceptions

None

Notice

See also

GetAlarmHistory()

TsRemote.method

Acquires alarm history.

Format

List<TsAlarm> GetAlarmHistory()

Argument

None

Return value

List<TsAlarm> Stores alarm history in the array of the TsAlarm object.

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

TsAlarm, GetCurrentAlarm()

GetCurrentAlarm()

TsRemote.method

Acquires the alarm that is currently occurring in the controller.

Format

List<TsAlarm> GetCurrentAlarm()

Argument

None

Return value

List<TsAlarm> Stores the alarm that is currently occurring in the controller in the array of the TsAlarm object.

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsAlarm

GetDIn()

TsRemote.method

Acquires the status of the specified input (Din).

Format

int GetDIn(int dino)

Argument

int dino

Specify a DIN number.

1~64: Standard input

101~164: Extended input

201~264: System input

301~364: FieldBus input

401~464: FieldBus input

Return value

int

Input status

0:OFF

1:ON

Exceptions

TCPNotConnect

Network is not connected

CommunicationError

Communication Error

ControllerNG

Command not accepted by controller

Notice

See also

GetDIO()

GetDIO()

TsRemote.method

Acquires the input/output status of the controller.

*Format***TsDIO GetDIO()***Argument*

None

*Return value***TsDIO**

Acquires the input (DIN) and output (DOOUT) statuses of the controller.

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsDIO,

GetDir()

TsRemote.method

Acquires the information of the files that are stored in the TS controller.

*Format***List<TsFileInfo> GetDir()***Argument*

None

*Return value***List<TsFileInfo>**

Stores the information of the file that is stored in the controller in the array of TsFileInfo object.

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsFileInfo

GetDout()

TsRemote.method

Acquires the alarm that is currently occurring in the TS controller.

Format

int GetDout(int dino)

Argument

int dino

Specify a DOUT number.

1~64: Standard output

101~164: Extended output

201~264: System output

301~364: FileBus output

401~464: FileBus output

Return value

int

Output status

0:OFF

1:ON

Exceptions

TCPNotConnect

Network is not connected

CommunicationError

Communication Error

ControllerNG

Command not accepted by controller

Notice

See also

GetDIn(), SetDout()

GetGlobalINT()

TsRemote.method

References the value of the integer type global variable of the program that is currently selected.

Format

int GetGlobalINT(string name)

Argument

string name	Name of the global variable For an array variable, specify including the element number.
--------------------	---

Return value

int	Value of the integer type global variable that was obtained
------------	---

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

SetGlobalINT(), GetGlobalPoint(), GetGlobalPAYLOAD(), GetGlobalREAL(), GetGlobalTRANS()

GetGlobalREAL()

TsRemote.method

References the value of the real number type global variable of the program that is currently selected.

Format

float GetGlobalREAL(string name)

Argument

string name	Name of the real number type global variable For an array variable, specify the element.
--------------------	---

Return value

float	Value of the real number type global variable that was acquired
--------------	---

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

GetGlobalINT(), GetGlobalPoint(), GetGlobalPAYLOAD(), SetGlobalREAL(), GetGlobalTRANS()

GetHexIn()

TsRemote.method

Acquires the status of input (DIN) with consecutive values from the specified output number.

Format

uint GetHexIn(int dino,int length)

Argument

int dino	First DIN number to be fetched consecutively
int length	Bit length for consecutive fetching

Return value

uint Value of DIN that was acquired
Example: DIN1=1,DIN2=0,DIN3=1,DIN4=0
The result of GetHexIn(1,4) is 5.

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

GetDin(), SetHexOut()

GetStatus()

TsRemote.method

Acquires the robot status.

*Format***TsStatus GetStatus()***Argument*

None

*Return value***TsStatus** Acquired status*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsStatus, GetStatusAll()

GetStatusMonitor()

TsRemote.method

Acquires the status monitor value of the robot.

*Format***TsStatusMonitor GetStatusMonitor()***Argument*

None

*Return value***TsStatusMonitor** Acquired monitor value*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

Use this command to acquire the status when WatchDog is inactive.
In the normal DLL use, acquire the status through the event by performing mutual monitoring by WatchDogStart().
Even if this command is issued, the controller status by the event cannot be acquired.

See also

WatchDogStart()

GetSystemVersion()

TsRemote.method

Acquires the system version of the TS controller.

Format

List<TsVersion> GetCTRLversion()

Argument

None

Return value

List<TsVersion> Stores the controller system version in the array of the TsVersion object.

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsVersion

ProgramBreak()

TsRemote.method

Stops the program execution for the robot immediately by BREAK.

*Format***void ProgramBreak()***Argument*

None

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

To detect ProgramBreak (stopping the operation) during execution of the Move*() method, perform exception processing by using try-catch for the Move() method.

See also

ProgramStop()

ProgramRun()

TsRemote.method

Executes the selected program.

*Format***void ProgramRun()***Argument*

None

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

ProgramStop(), ProgramBreak()

ProgramSelect()

TsRemote.method

Selects a program.

Format

void ProgramSelect(string filename)

Argument

string filename Name of the file to be selected

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

GetDir(), ProgramRun()

ProgramStop()

TsRemote.method

Stops the program that is being executed.

*Format***void ProgramStop()***Argument*

None

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

To detect ProgramStop (stopping the operation) during execution of the Move*() method, perform exception processing by using try-catch for the Move() method.

See also

ProgramRun(), ProgramBreak()

ReceiveFile()

TsRemote.method

Receives the file in the robot controller in the personal computer.

Format

void ReceiveFile(string filename, string fullpass)

Argument

string filename	Name of the file to be received
string fullpass	File storage location and name under which the received file is stored

Return value

None

Exceptions

NotfoundDirectory	Directory is not found in PC
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

SendFile(), GetDir()

ResetALARM()

TsRemote.method

Performs alarm reset.

Format

void ResetAlarm()

Argument

None

Return value

None

Exceptions

TCPNotConnect Network is not connected
CommunicationError Communication Error
ControllerNG Command not accepted by controller

Notice

See also

ResetCYCLE(), ResetPRG(), ResetMove(), ResetSELECT(), ResetSIGNAL(), ResetSTEP()

ResetCYCLE()

TsRemote.method

Sets the program to the executable state from the RCYCLE by resetting the execution environment except for the global variables of the selected program.

*Format***void ResetCYCLE()***Argument*

None

Return value

None

Exceptions

TCPNotConnect Network is not connected
CommunicationError Communication Error
ControllerNG Command not accepted by controller

*Notice**See also*

ResetALARM(), ResetPRG(), ResetMove(), ResetSELECT(), ResetSIGNAL(), ResetSTEP()

ResetPRG()

TsRemote.method

Resets the execution environment of the selected program.

*Format***void ResetPRG()***Argument*

None

Return value

None

Exceptions

TCPNotConnect Network is not connected
CommunicationError Communication Error
ControllerNG Command not accepted by controller

*Notice**See also*

ResetALARM(), ResetCYCLE(), ResetMove(), ResetSELECT(), ResetSIGNAL(), ResetSTEP()

ResetMove()

TsRemote.method

Enables the operation of the Move*() method by clearing the DLL internal flag (Move Stop) for the stopping of the robot operation,

*Format***void ResetMove()***Argument*

None

Return value

None

Exceptions

None

Notice

When an event occurs due to the EmergencyStop, SafetyStop, StopCommand, or BreakCommand during the mutual monitoring by WatchDogStart(), the DLL internal flag (MoveStop) is set to ON.

If the Move*() or DirectDo() method is executed while the internal flag (MoveStop) is ON, MoveStop exception occurs, thereby inhibiting the robot operation. ResetMove() enables the robot operation by resetting the internal flag. Reset the state by checking with the user through a dialog and so on.

See also

WatchDogStart(), TsStatusMonitor, Move(), Movea(), Movej(), Moves(),

ResetSELECT()

TsRemote.method

Resets the program selection.

*Format***void ResetSELECT()***Argument*

None

Return value

None

Exceptions

TCPNotConnect Network is not connected
CommunicationError Communication Error
ControllerNG Command not accepted by controller

*Notice**See also*

ResetALARM(), ResetCYCLE(), ResetPRG(), ResetMove(), ResetSIGNAL(), ResetSTEP()

ResetSIGNAL()

TsRemote.method

Resets the output signal of the robot.

*Format***void ResetSIGNAL()***Argument*

None

Return value

None

Exceptions

TCPNotConnect Network is not connected
CommunicationError Communication Error
ControllerNG Command not accepted by controller

*Notice**See also*

ResetALARM(), ResetCYCLE(), ResetPRG(), ResetMove(), ResetSIGNAL(), ResetSTEP()

ResetSTEP()

TsRemote.method

Enables execution of the selected program by resetting the execution environment except for the global variables of the program.

*Format***void ResetSTEP()***Argument*

None

Return value

None

Exceptions

TCPNotConnect Network is not connected
CommunicationError Communication Error
ControllerNG Command not accepted by controller

*Notice**See also*

ResetALARM(), ResetCYCLE(), ResetPRG(), ResetMove(), ResetSELECT(), ResetSIGNAL()

SendFile()

TsRemote.method

Sends the file in the personal computer to the robot controller.

Format

void SendFile(string filename, string fullpass)

Argument

string filename	Name under which the file that is sent is to be stored in the controller
string fullpass	Directory and file name to be sent

Return value

None

Exceptions

NotFondFile	File is not found
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

ReceiveFile()

ServoOff()

TsRemote.method

Sets the servo to OFF.

*Format***void ServoOff()***Argument*

None

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

ServoOn()

ServoOn()

TsRemote.method

Sets the servo to ON.

*Format***void ServoOn()***Argument*

None

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

ServoOff()

SetComPort()

TsRemote.method

Sets the parameters of the COM port.

Format

bool SetComPort(string portNo,int bps,int bit, int parity,int stop)

Argument

string portNo	COM port number
int32 bps	Transfer speed
int bit	Number of bits per character (7 or 8)
int parity	Parity (0: None, 1: Even number, 2: Odd number)
int stop	Stop bit (1: Stop bit 1)

Return value

bool	true: Successful false: Failed
-------------	-----------------------------------

Exceptions

IllegalStopbit	Illegal stopbits.Should be 0,1,or 2
IllegalDatabits	Illegal databits.Should be 7 or 8
IllegalBps	Illegal baud rate. Should be between 1200 to 115200
IllegalPortNo	Illegal port No.

Notice

SetComPort() is used for function extension and currently the function has not been installed.

See also

SetDout()

TsRemote.method

Acquires the input (DOUT) status.

Format

void SetDout(int dono, int setdata)

Argument

int dono	DOUT number
int setdata	0:OFF 1:ON

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

GetDout()

SetGlobalINT()

TsRemote.method

Sets the value of the integer type global variable of the program that is currently selected.

Format

void SetGlobalINT(string name, int restore,int val)

Argument

string name	Name of the global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0 : Not performed 1 : Performed
Int val	Value to be set

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

GetGlobalINT(), SetGlobalPoint(), SetGlobalPAYLOAD(), SetGlobalREAL(), SetGlobalTRANS()

SetGlobalREAL()

TsRemote.method

Sets the value of the program real number type global variable of the program that is currently selected.

Format

```
void SetGlobalREAL(string name,int restore,double val )
```

Argument

string name	Name of the global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0 : Not performed 1 : Performed
double val	Value to be set

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

SetGlobalINT(), SetGlobalPoint(), SetGlobalPAYLOAD(), GetGlobalREAL(), SetGlobalTRANS()

SetHexOut()

TsRemote.method

Sets consecutive numbers starting from the specified output numbers.

Format

void SetHexOut(int dono,int length,uint value)

Argument

Int dono	First DOUT number to be set consecutively
Int length	Bit length to be fetched consecutively (1~32)
Int value	Set value

Example: The result of SetHexOut(1,4,5) is DOUT1=1, DOUT2=0, DOUT3=1, DOUT4=0.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

GetHexIn()

SetIPAddr()

TsRemote.method

Sets Ethernet connection.

Format

bool SetIPAddr(int openmode,string destAddr,int destPort,int srcPort)

Argument

Int openmode	0: Server(not installed) 1: Client
string destAddr	IP address of the connection destination (robot controller) Use a character string such as "192.168.0.100".
Int destPort	Port number of the connection destination (robot controller)
Int srcPort	Port number of the connection source (personal computer)

Return value

bool	true: Successful false: Failed
-------------	-----------------------------------

Exceptions

IllegalIPAddress	Illegal IP Address
------------------	--------------------

Notice

The Server function is not currently supported.

See also

Connect()

SetOVRD()

TsRemote.method

Sets override.

*Format***void SetOVRD(int override)***Argument***Int** override Override setting value (1 to 100)*Return value*

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

The override that is set is inherited even if the master mode is changed.

See also

GetOVRD()

SetRunMode()

TsRemote.method

Sets RUNMODE (run mode).

Format

```
void SetRunMode(RunMode mode)
void SetRunMode(int mode)
```

Argument

RunMode mode	Specify RUNMODE with RunMode enumeration.
Int mode	Set RUNMODE using one of the following values. 0: The robot operation mode is CONT. 1: The robot operation mode is CYCLE. 2: The robot operation mode is STEP. 3: The robot operation mode is SEGMENT.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

WatchDogStart()

TsRemote.method

Starts mutual monitoring between the controller and PC.

Format

void WatchDogStart(int intervalTime,int timeOut,int alarmLevel)

Argument

Int intervalTime	Communication interval of mutual monitoring (100 to 1000msec)
Int timeOut	Time to produce time-out alarm in mutual monitoring (Set time more than twice intervalTime)
Int alarmLevel	Alarm that is issued when a problem occurs in mutual monitoring 0: No alarm 1: 1-200 Host Port Time Out Message only (The robot does not stop) 2: 2-137 Host Port Time Out Stops when the operation that is currently performed reaches the target point. 8: 8-352 Host Port Time Out Stops immediately.

Return value

None

Exceptions

IllegalWatchDogInterval	Illegal watchdog monitoring interval.Should be between 0 to 9999
IllegalAlarmLevel	Illegal alarm level.Should be 0 or 1 or 2 or 8
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

To execute Move(), Movea(), Movej(), Moves() and DirectDo() method, mutual monitoring must be enabled.

See also

WatchDogStop()

WatchDogStop()

TsRemote.method

Stops mutual monitoring between the controller and PC.

*Format***void WatchDogStop()***Argument*

None

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

WatchDogStart()

3.6 TsRemoteS Class property

MvConfig

TsRemoteS.property

Specifies/acquires the configuration (posture) of the SCARA robot operation that is instructed by Move(), Moves(), Movej(), and Movea().

Format

MvConfigs

Value

enum ConfigS

ConfigS.FREE: Sets the configuration (posture) of SCARA robot to FREE.
 ConfigS.LEFTY: Sets the configuration (posture) of SCARA robot to LEFTY.
 ConfigS.RIGHTY: Sets the configuration (posture) of SCARA robot to RIGHTY.

Notice

The MvConfig property value has the priority over the configuration that is specified in the Move*() method.
 To specify configuration in the Move*() operation instruction, set the MvConfig property value to FREE.

See also

MvTool

TsRemoteS.property

Tool coordinate of the SCARA robot operation that is instructed by Move(), Moves(), Movej(), and Movea().

Format

MvTool

Value

TsTransS

Specify the set value with TsTransS.

Notice

See also

MvAccel, MvSpeed, MvConfig, Move(), Moves(), Movej(), Movea()

MvWork

TsRemoteS.property

Work coordinate of the SCARA robot operation that is instructed by Move(), Moves(), Movej(), and Movea().

Format

MvWork

Value

TsTransS

Specify the set value with TsTransS.

Notice

See also

MvAccel, MvSpeed, MvConfig, Move(), Moves(), Movej(), Movea()

3.7 TsRemoteS Class Method

GetGlobalPOINT()

TsRemoteS.method

References the value of the point type global variable of the program that is currently selected.

Format

TsPointS GetGlobalPOINT(string name)

Argument

string name	Name of the point type global variable (teaching point) For an array variable, specify the element.
--------------------	--

Return value

TsPointS	Value of the point type global variable that was acquired
-----------------	---

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

TsPointS, GetGlobalINT(), SetGlobalPoint(), GetGlobalPAYLOAD(),GetGlobalREAL(),
GetGlobalTRANS()

GetGlobalPAYLOAD()

TsRemoteS.method

References the value of the load type global variable of the program that is currently selected.

Format

TsPayloadS GetGlobalPAYLOAD(string name)

Argument

string name	Name of the load type global variable For an array variable, specify the element.
--------------------	--

Return value

TsPayloadS	Value of the load type global variable that was acquired
-------------------	--

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

TsPayload, GetGlobalINT(), GetGlobalPoint(), SetGlobalPAYLOAD(), GetGlobalREAL(),
GetGlobalTRANS()

GetGlobalTRANS()

TsRemoteS.method

References the value of the coordinate type global variable of the program that is currently selected.

Format

TsTransS GetGlobalTRANS(string name)

Argument

string name	Name of the standard type global variable For an array variable, specify the element.
--------------------	--

Return value

TsTransS	Value of the coordinate type global variable that was acquired
-----------------	--

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

TsTransS, GetGlobalINT(), GetGlobalPoint(), GetGlobalPAYLOAD(), GetGlobalREAL(),
SetGlobalTRANS()

GetPsnCmdJoint()

TsRemoteS.method

Acquires the current robot position (command) in terms of the joint angle.

*Format***TsJointS GetPsnCmdJoint()***Argument*

None

*Return value***TsJointS** Joint angle of SCARA robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkJoint(), GetPsnFbkWork(), GetPsnFbkWorld()

GetPsnCmdWork()

TsRemoteS.method

Acquires the current robot position (command position) in terms of the work coordinate position.

*Format***TsPointS GetPsnCmdWork()***Argument*

None

*Return value***TsPointS** Work coordinate point of SCARA robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointS, GetPsnCmdJoint(), GetPsnCmdWorld(), GetPsnFbkJoint(), GetPsnFbkWork()
GetPsnFbkWorld()

GetPsnCmdWorld()

TsRemoteS.method

Acquires the current robot point (command position) in terms of the world coordinate position.

*Format***TsPointS GetPsnCmdWorld()***Argument*

None

*Return value***TsPointS** World coordinate position of SCARA robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointS, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnFbkJoint(), GetPsnFbkWork()
GetPsnFbkWorld()

GetPsnFbkJoint()

TsRemoteS.method

Acquires the current robot position (feedback) in terms of the joint angle.

Format

TsJointS GetPsnFbkJoint()

Argument

None

Return value

TsJointS Joint angle of SCARA robot (calculated from the motor angle)

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

TsJointS, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkWork()
GetPsnFbkWorld()

GetPsnFbkWork()

TsRemoteS.method

Acquires the current robot position (feedback) in terms of the work coordinate position.

*Format***TsPointS GetPsnFbkWork()***Argument*

None

*Return value***TsPointS** Work coordinate position of SCARA robot (calculated from the motor angle)*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointS, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkJoint()
GetPsnFbkWorld()

GetPsnFbkWorld()

TsRemoteS.method

Acquires the current robot position (feedback) in terms of the world coordinate position.

*Format***TsPointS GetPsnFbkWorld()***Argument*

None

*Return value***TsPointS** World coordinate position of SCARA robot (calculated from the motor angle)*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointS, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkJoint()
GetPsnFbkWork()

GetStatusAll()

TsRemoteS.method

Acquires all the statuses of the SCARA robot.

Format

TsStatusAllS **GetStatusAll()**

Argument

None

Return value

TsStatusAllS Acquired statuses of SCARA robot

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsStatusAllS , GetStatus()

GetVelocJoint()

TsRemoteS.method

Acquires the current robot command speed in terms of the joint angle.

*Format***TsJointS GetVelocJoint()***Argument*

None

*Return value***TsJointS** Speed of each joint of SCARA robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsJoint, GetVelocWorld()

GetVelocWorld()

TsRemoteS.method

Acquires the current command speed of the robot in terms of the speed of the world coordinate elements.

*Format***TsPointS GetVelocWorld()***Argument*

None

*Return value***TsPointS** Speed of SCARA robot (calculated based on the X, Y, Z, and C components)*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointS, GetVelocJoint()

Move()

TsRemoteS.method

Moves the SCARA robot to the specified point.

Format

```

Move(double x, double y, double z, double c, double t, int config)
Move(TsPointS point)
Move(TsPointS point, double px, double py, double pz, double pc, double pt)
Move(TsPointS point, TsPointS offset)
Move(string pointName)
Move(string pointName, double px, double py, double pz, double pc, double pt)
Move(string pointName, TsPointS offset)

```

Argument

double x	X coordinate target point
double y	Y coordinate target point
double z	Z coordinate target point
double c	C coordinate target point
double t	T coordinate target point
TsPointS point	Target point
double px	Offset value to be added to the X coordinate target point
double py	Offset value to be added to the Y coordinate target point
double pz	Offset value to be added to the Z coordinate target point
double pc	Offset value to be added to the C coordinate target point
double pt	Offset value to be added to the T coordinate target point
TsPointS offset	Offset value to be added to the target point
string pointName	Name of the point that is registered in the selected program

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
MoveBreak	BREAK signal during the Move command
MoveTimeout	Move command execution timed out
CannotDoMove	Move command can not be executed, because of watch dog not running
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

Move() cannot be executed unless monitoring is performed by the WatchDogStart() method.
 To detect BREAK or STOP (stopping the operation) during execution of Move(), perform exception processing by using try-catch.

See also

WatchDogStart(), ResetMove(), Movea(), Movej(), Moves(), TsPoint

Movea()

TsRemoteS.method

Moves the joint axis of the SCARA robot to the specified position.

Format

void Movea(int joint_no,double position)

Argument

int joint_no	Joint number(1-5) to be moved
double position	Target position (angle)

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
MoveBreak	BREAK signal during the Move command
MoveTimeout	Move command execution timed out
CannotDoMove	Move command can not be executed, because of watch dog not running
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

Move() cannot be executed unless monitoring is performed by the WatchDogStart() method.
To detect BREAK or STOP (stopping the operation) during execution of Move(), perform exception processing by using try-catch.

See also

WatchDogStart(), ResetMove(), Movea(), Movej(), Moves()

Movej()

TsRemoteS.method

Moves the SCARA robot to the specified point with an arch motion.

Format

```
void Movej(double x, double y, double z, double c, double t, int config,
           double l1, double l2, double l3)
void Movej(TsPointS point, double l1, double l2, double l3)
void Movej(TsPointS point, TsPointS offset, double l1, double l2, double l3)
void Movej(string pointName, double l1, double l2, double l3)
void Movej(string pointName, TsPointS offset, double l1, double l2, double l3)
```

Argument

double x	X coordinate target point
double y	Y coordinate target point
double z	Z coordinate target point
double c	C coordinate target point
double t	T coordinate target point
TsPointS point	Target point
double l1	Refuge movement distance
double l2	Single-axis movement distance 1
double l3	Single-axis movement distance 2
TsPointS offset	Offset value to be added to the target point
string pointName	Name of the point that is registered in the selected program

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
MoveBreak	BREAK signal during the Move command
MoveTimeout	Move command execution timed out
CannotDoMove	Move command can not be executed, because of watch dog not running
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

Move() cannot be executed unless monitoring is performed by the WatchDogStart() method.
To detect BREAK or STOP (stopping the operation) during execution of Move(), perform exception processing by using try-catch.

See also

WatchDogStart(), Move(), ResetMove(), Movea(), Moves(), TsPointS

Moves()

TsRemoteS.method

Moves the SCARA robot to the specified point with linear interpolation.

Format

```
void Moves(double x, double y, double z, double c, double t, int config)
void Moves(TsPointS point)
void Moves(TsPointS point,double px, double py, double pz, double pc, double pt)
void Moves(TsPointS point,tsPointS offset)
void Moves(string pointName)
void Moves(string pointName,double px, double py, double pz, double pc, double pt)
void Moves(string pointName,TsPointS offset)
```

Argument

double x	X coordinate target point
double y	Y coordinate target point
double z	Z coordinate target point
double c	C coordinate target point
double t	T coordinate target point
TsPointS point	Target point
double px	Offset value to be added to the X coordinate target point
double py	Offset value to be added to the Y coordinate target point
double pz	Offset value to be added to the Z coordinate target point
double pc	Offset value to be added to the C coordinate target point
double pt	Offset value to be added to the T coordinate target point
TsPointS offset	Offset value to be added to the target point
string pointName	Name of the point that is registered in the selected program

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
MoveBreak	BREAK signal during the Move command
MoveTimeout	Move command execution timed out
CannotDoMove	Move command can not be executed, because of watch dog not running
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

Move() cannot be executed unless monitoring is performed by the WatchDogStart() method.
To detect BREAK or STOP (stopping the operation) during execution of Move(), perform exception processing by using try-catch.

See also

WatchDogStart(), Move(), ResetMove(), Movea(), Movej(), TsPointS

SetGain()

TsRemoteS.method

Sets the gain of each axis of the robot (servo control) to ON/OFF.

Format

void SetGain(int j1,int j2, int j3, int j4, int j5)

Argument

Int j1	Servo gain of the 1 st joint (0:OFF 1:ON)
Int j2	Servo gain of the 2 nd joint (0:OFF 1:ON)
Int j3	Servo gain of the 3 rd joint (0:OFF 1:ON)
Int j4	Servo gain of the 4 th joint (0:OFF 1:ON)
Int j5	Servo gain of the 5 th joint (0:OFF 1:ON)

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

SetGain() changes the gain at the following operation (Move*()).

See also

SetGlobalPOINT()

TsRemoteS.method

Sets the value of the program point type global variable of the program that is currently selected.

Format

```
void SetGlobalPoint(string name,int restore,double x, double y, double z,
                    double c, double t, ConfigS config)
voidSetGlobalPoint(string name,int restore,TsPointS point)
```

Argument

string name	Name of the global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0 : Not performed 1 : Performed
double x	Value set for X
double y	Value set for Y
double z	Value set for Z
double c	Value set for C
double t	Value set for T
ConfigS config	Value set for Config FREE: Free LEFTY: Left hand system RIGHTY: Right hand system
TsPointS point	Specify the set value with TsPointS.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

TsPointS, SetGlobalINT(), GetGlobalPoint(), SetGlobalPAYLOAD(), SetGlobalREAL()
SetGlobalTRANS()

SetGlobalPYLOAD()

TsRemoteS.method

Sets the value of the program load type global variable of the program that is currently selected.

Format

SetGlobalPYLOAD(string name,int restore,double m,double l)
SetGlobalPYLOAD(string name,int restore,TsPayloadS payload)

Argument

string name	Name of the global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0: Not performed 1: Performed
double m	Weight to be set(Kg)
double l	Offset value to be set(mm)
TsPayloadS payload	Specify the set value with TsPayloadS.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

SetGlobalINT(), SetGlobalPoint(), GetGlobalPAYLOAD(), SetGlobalREAL(), SetGlobalTRANS()

SetGlobalTRANS()

TsRemoteS.method

Sets the value of the coordinate type global variable of the program that is currently selected.

Format

```
void SetGlobalTRANS(string name,int restore,double x,double y,double z,double c)
void SetGlobalTRANS(string name,int restore,TsTransS trans)
```

Argument

string name	Name of the coordinate type global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0 : Not performed 1 : Performed
double x	Value set for X
double y	Value set for Y
double z	Value set for Z
double c	Value set for C
TsTransS trans	Specify the set value with TsTransS.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

SetGlobalINT(), SetGlobalPoint(), SetGlobalPAYLOAD(), SetGlobalREAL(), GetGlobalTRANS()

SetPayload()

TsRemoteS.method

Sets the load data of the SCARA robot.

Format

```
void SetPayload(double m, double l)
void SetPayload(TsPayloadS payload)
```

Argument

double m	Load data (Weight Kg)
double l	Load data (Offset of center of gravity mm)
TsPayloadS payload	Specify the set value with TsPayLoadS.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

The load information that is set is inherited even if the master mode is changed.
To clear load data, specify SetPayload(0,0).

See also

SetTransTool()

TsRemoteS.method

Sets the tool coordinate.

Format

```
void SetTransTool(string name) *1  
void SetTransTool(string name,double x,double y,double z,double c) *2  
void SetTransTool(string name,TsTransS tool) *2
```

Format

string name	*1 Specify the name of the coordinate type variable that is registered in the program file that is selected by the controller.
	*2 Names for the x,y,z, and c values that are set for convenience (displayed in TP)
double x	X set value
double y	Y set value
double z	Z set value
double c	C set value
TsTransS tool	Specify the set value with TsTrans.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

The tool coordinate that is set is inherited even if the master mode is changed.

See also

SetTransWork()

TsRemoteS.method

Sets the work coordinate.

Format

```
void SetTransWork(string name)*1  
void SetTransWork(string name,double x,double y,double z,double c)*2  
void SetTransWork(string name,TsTransS work)*2
```

Argument

string name	*1 Specify the name of the coordinate type variable that is registered in the program file that is selected by the controller.
	*2 Names for the x,y,z, and c values that are set for convenience (displayed in TP)
double x	X set value
double y	Y set value
double z	Z set value
double c	C set value
TsTransS work	Specify the set value with TsTrans.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

The work coordinate that is set is inherited even if the master mode is changed.

See also

3.8 TsRemoteV Class property

MvConfig

TsRemoteV.property

Specifies/acquires the configuration (posture) of the 6-AXIS robot operation that is instructed by Move(), Moves(), Movej(), and Movea().

*Format***MvConfig***Value***enum ConfigV** 6-Axis robot configuration.*Notice*

The MvConfig property value has the priority over the configuration that is specified in the Move*() method.

To specify configuration in the Move*() operation instruction, set the MvConfig property value to "xxxxxx" (0).

See also

enum CongigV

MvTool

TsRemoteV.property

Tool coordinate of the 6-AXIS robot operation that is instructed by Move(), Moves(), Movej(), and Movea().

*Format***MvTool***Value***TsTransV** Specify the set value with TsTransV.*Notice**See also*

MvAccel, MvSpeed, MvConfig, Move(), Moves(), Movej(), Movea()

MvWork

TsRemoteV.property

Work coordinate of the 6-AXIS robot operation that is instructed by Move(), Moves(), Movej(), and Movea().

*Format***MvWork***Value***TsTransV** Specify the set value with TsTransV.*Notice**See also*

MvAccel, MvSpeed, MvConfig, Move(), Moves(), Movej(), Movea()

3.9 TsRemoteS Class Method

GetGlobalPOINT()

TsRemoteV.method

References the value of the point type global variable of the program that is currently selected.

*Format***TsPointV GetGlobalPOINT(string name)***Argument*

string name	Name of the point type global variable (teaching point) For an array variable, specify the element.
--------------------	--

Return value

TsPointV	Value of the point type global variable that was acquired
-----------------	---

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

*Notice**See also*

TsPointS, GetGlobalINT(), SetGlobalPoint(), GetGlobalPAYLOAD(),GetGlobalREAL(),
GetGlobalTRANS()

GetGlobalPAYLOAD()

TsRemoteV.method

References the value of the load type global variable of the program that is currently selected.

Format

TsPayloadV GetGlobalPAYLOAD(string name)

Argument

string name	Name of the load type global variable For an array variable, specify the element.
--------------------	--

Return value

TsPayloadV	Value of the load type global variable that was acquired
-------------------	--

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

TsPayload, GetGlobalINT(), GetGlobalPoint(), SetGlobalPAYLOAD(), GetGlobalREAL(),
GetGlobalTRANS()

GetGlobalTRANS()

TsRemoteV.method

References the value of the coordinate type global variable of the program that is currently selected.

Format

TsTransV GetGlobalTRANS(string name)

Argument

string name	Name of the standard type global variable For an array variable, specify the element.
--------------------	--

Return value

TsTransS	Value of the coordinate type global variable that was acquired
-----------------	--

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

See also

TsTransV, GetGlobalINT(), GetGlobalPoint(), GetGlobalPAYLOAD(), GetGlobalREAL(), SetGlobalTRANS()

GetPsnCmdJoint()

TsRemoteV.method

Acquires the current robot position (command) in terms of the joint angle.

*Format***TsJointS GetPsnCmdJoint()***Argument*

None

*Return value***TsJointV** Joint angle of 6-AXIS robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkJoint(), GetPsnFbkWork(), GetPsnFbkWorld()

GetPsnCmdWork()

TsRemoteV.method

Acquires the current robot position (command position) in terms of the work coordinate position.

*Format***TsPointV GetPsnCmdWork()***Argument*

None

*Return value***TsPointV** Work coordinate point of 6-AXIS robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointV, GetPsnCmdJoint(), GetPsnCmdWorld(), GetPsnFbkJoint(), GetPsnFbkWork()
GetPsnFbkWorld()

GetPsnCmdWorld()

TsRemoteV.method

Acquires the current robot point (command position) in terms of the world coordinate position.

*Format***TsPointV GetPsnCmdWorld()***Argument*

None

*Return value***TsPointV** World coordinate position of 6-AXIS robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointV, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnFbkJoint(), GetPsnFbkWork()
GetPsnFbkWorld()

GetPsnFbkJoint()

TsRemoteV.method

Acquires the current robot position (feedback) in terms of the joint angle.

Format

TsJointV GetPsnFbkJoint()

Argument

None

Return value

TsJointV Joint angle of 6-AXIS robot (calculated from the motor angle)

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

TsJointV, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkWork()
GetPsnFbkWorld()

GetPsnFbkWork()

TsRemoteV.method

Acquires the current robot position (feedback) in terms of the work coordinate position.

*Format***TsPointV GetPsnFbkWork()***Argument*

None

*Return value***TsPointV** Work coordinate position of 6-AXIS robot (calculated from the motor angle)*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointV, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkJoint()
GetPsnFbkWorld()

GetPsnFbkWorld()

TsRemoteV.method

Acquires the current robot position (feedback) in terms of the world coordinate position.

*Format***TsPointV GetPsnFbkWorld()***Argument*

None

*Return value***TsPointV** World coordinate position of 6-AXIS robot (calculated from the motor angle)*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointV, GetPsnCmdJoint(), GetPsnCmdWork(), GetPsnCmdWorld(), GetPsnFbkJoint()
GetPsnFbkWork()

GetStatusAll()

TsRemoteV.method

Acquires all the statuses of the 6-AXIS robot.

Format

TsStatusAllV GetStatusAll()

Argument

None

Return value

TsStatusAllV Acquired statuses(6-AXIS)

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsStatusAllV , GetStatus()

GetVelocJoint()

TsRemoteV.method

Acquires the current robot command speed in terms of the joint angle.

*Format***TsJointV GetVelocJoint()***Argument*

None

*Return value***TsJointV** Speed of each joint of 6-AXIS robot*Exceptions*

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsJoint, GetVelocWorld()

GetVelocWorld()

TsRemoteV.method

Acquires the current command speed of the robot in terms of the speed of the world coordinate elements.

*Format***TsPointV GetVelocWorld()***Argument*

None

*Return value***TsPointV**

Speed of 6-AXIS robot
(calculated based on the X, Y, Z,A,B and C components)

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

*Notice**See also*

TsPointV, GetVelocJoint()

Move()

TsRemoteV.method

Moves the 6-AXIS robot to the specified point.

Format

```

Move(double x, double y, double z, double a, double b, double c,
      double u, double v, ConfigV config)
Move(double x, double y, double z, double a, double b, double c,
      double u, double v, ConfigVpart config)

Move(TsPointV point)
Move(TsPointV point, double px, double py, double pz,
      double pa, double pb, double pc, double pu, double pv)
Move(TsPointV point, TsPointV offset)
Move(string pointName)
Move(string pointName, double px, double py, double pz,
      double pa, double pb, double pc, double pu, double pv)
Move(string pointName, TsPointV offset)

```

Argument

double x	X coordinate target point
double y	Y coordinate target point
double z	Z coordinate target point
double a	A coordinate target point
double b	B coordinate target point
double c	C coordinate target point
double u	U coordinate target point
double v	V coordinate target point
ConfigV config	configuration(enum)
ConfigVpart config	configuration(class)
TsPointV point	Target point
double px	Offset value to be added to the X coordinate target point
double py	Offset value to be added to the Y coordinate target point
double pz	Offset value to be added to the Z coordinate target point
double pa	Offset value to be added to the A coordinate target point
double pb	Offset value to be added to the B coordinate target point
double pc	Offset value to be added to the C coordinate target point
double pu	Offset value to be added to the U coordinate target point
double pv	Offset value to be added to the V coordinate target point
TsPointV offset	Offset value to be added to the target point
string pointName	Name of the point that is registered in the selected program

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
MoveBreak	BREAK signal during the Move command
MoveTimeout	Move command execution timed out
CannotDoMove	Move command can not be executed, because of watch dog not running
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

Move() cannot be executed unless monitoring is performed by the WatchDogStart() method.
To detect BREAK or STOP (stopping the operation) during execution of Move(), perform exception processing by using try-catch.

See also

WatchDogStart(), ResetMove(), Movea(), Movej(), Moves(), TsPoint

Movea()

TsRemoteV.method

Moves the joint axis of the 6-AXIS robot to the specified position.

Format

void Movea(int joint_no,double position)

Argument

int joint_no	Joint number(1-8) to be moved
double position	Target position (angle)

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
MoveBreak	BREAK signal during the Move command
MoveTimeout	Move command execution timed out
CannotDoMove	Move command can not be executed, because of watch dog not running
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

Move() cannot be executed unless monitoring is performed by the WatchDogStart() method.
To detect BREAK or STOP (stopping the operation) during execution of Move(), perform exception processing by using try-catch.

See also

WatchDogStart(), ResetMove(), Movea(), Movej(), Moves()

Moves()

TsRemoteV.method

Moves the 6-AXIS robot to the specified point with linear interpolation.

Format

```
void Moves(double x, double y, double z, double a, double b, double c,
           double u, double v, ConfigV config)
void Moves(double x, double y, double z, double a, double b, double c,
           double u, double v, ConfigVpart config)

void Moves(TsPointV point)
void Moves(TsPointV point, double px, double py, double pz,
           double pa, double pb, double pc, double pu, double pv)
void Moves(TsPointV point, TsPointV offset)
void Moves(string pointName)
void Moves(string pointName, double px, double py, double pz,
           double pa, double pb, double pc, double pu, double pv)
void Moves(string pointName, TsPointV offset)
```

Argument

double x	X coordinate target point
double y	Y coordinate target point
double z	Z coordinate target point
double a	A coordinate target point
double b	B coordinate target point
double c	C coordinate target point
double u	U coordinate target point
double v	V coordinate target point
ConfigV config	configuration(enum)
ConfigVpart config	configuration(class)
TsPointV point	Target point
double px	Offset value to be added to the X coordinate target point
double py	Offset value to be added to the Y coordinate target point
double pz	Offset value to be added to the Z coordinate target point
double pa	Offset value to be added to the A coordinate target point
double pb	Offset value to be added to the B coordinate target point
double pc	Offset value to be added to the C coordinate target point
double pu	Offset value to be added to the U coordinate target point
double pv	Offset value to be added to the V coordinate target point
TsPointV offset	Offset value to be added to the target point
string pointName	Name of the point that is registered in the selected program

Return value

None

Exceptions

MoveStop	STOP signal during the Move command
MoveBreak	BREAK signal during the Move command
MoveTimeout	Move command execution timed out
CannotDoMove	Move command can not be executed, because of watch dog not running
TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

Move() cannot be executed unless monitoring is performed by the WatchDogStart() method.
To detect BREAK or STOP (stopping the operation) during execution of Move(), perform exception processing by using try-catch.

See also

WatchDogStart(), Move(), ResetMove(), Movea(), Movej(), TsPointV

SetGain()

TsRemoteV.method

Sets the gain of each axis of the robot (servo control) to ON/OFF.

Format

```
void SetGain(int j1,int j2, int j3, int j4, int j5, int j6, int j7, int j8)
```

Argument

Int j1	Servo gain of the 1 st joint (0:OFF 1:ON)
Int j2	Servo gain of the 2 nd joint (0:OFF 1:ON)
Int j3	Servo gain of the 3 rd joint (0:OFF 1:ON)
Int j4	Servo gain of the 4 th joint (0:OFF 1:ON)
Int j5	Servo gain of the 5 th joint (0:OFF 1:ON)
Int j6	Servo gain of the 6 th joint (0:OFF 1:ON)
Int j7	Servo gain of the 7 th joint (0:OFF 1:ON)
Int j8	Servo gain of the 8 th joint (0:OFF 1:ON)

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

SetGain() changes the gain at the following operation (Move*()).

See also

SetGlobalPAYLOAD()

TsRemoteV.method

Sets the value of the program load type global variable of the program that is currently selected.

Format

```
SetGlobalPYLOAD(string name,int restore,double m,  
                double Gx, double Gy, double Gz, double Lx, double Ly, double Lz)  
SetGlobalPYLOAD(string name,int restore,TsPayloadV payload)
```

Argument

string name	Name of the global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0: Not performed 1: Performed
double m	Weight to be set(Kg)
double M	Weight (kg)
double Lx	Offset of the center of gravity (mm)
double Ly	Offset of the center of gravity (mm)
double Lz	Offset of the center of gravity (mm)
double Gx	Work size(mm)
double Gy	Work size(mm)
double Gz	Work size(mm)
TsPayloadV payload	Specify the set value with TsPayloadV.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

SetGlobalINT(), SetGlobalPoint(), GetGlobalPAYLOAD(), SetGlobalREAL(), SetGlobalTRANS()

SetGlobalPOINT()

TsRemoteV.method

Sets the value of the program point type global variable of the program that is currently selected.

Format

```
void SetGlobalPoint(string name,int restore,double x, double y, double z,
    double a, double b, double c, double u, double v, ConfigVpart config)
voidSetGlobalPoint(string name,int restore,TsPointV point)
```

Argument

string name	Name of the global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0 : Not performed 1 : Performed
double x	Value set for X
double y	Value set for Y
double z	Value set for Z
double a	Value set for A
double b	Value set for B
double c	Value set for C
double u	Value set for U
double v	Value set for V
ConfigVpart config	Value set for ConfigVpart class
TsPointV point	Specify the set value with TsPointV.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

TsPointV, SetGlobalINT(), GetGlobalPoint(), SetGlobalPAYLOAD(), SetGlobalREAL()
SetGlobalTRANS()

SetGlobalTRANS()

TsRemoteV.method

Sets the value of the coordinate type global variable of the program that is currently selected.

Format

```
void SetGlobalTRANS(string name,int restore,double x,double y,double z,  
                    double a , double b ,double c)  
void SetGlobalTRANS(string name,int restore,TsTransV trans)
```

Argument

string name	Name of the coordinate type global variable When specifying an array variable, include the element number also.
Int restore	Set whether RESTORE (change of the initial value of the program file) is performed or not. 0 : Not performed 1 : Performed
double x	Value set for X
double y	Value set for Y
double z	Value set for Z
double a	Value set for A
double b	Value set for B
double c	Value set for C
TsTransV trans	Specify the set value with TsTrans.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller

Notice

See also

SetGlobalINT(), SetGlobalPoint(), SetGlobalPAYLOAD(), SetGlobalREAL(), GetGlobalTRANS()

SetPayload()

TsRemoteV.method

Sets the load data of the 6-AXIS robot.

Format

```
void SetPayload (double m,double Gx, double Gy, double Gz,  
                double Lx, double Ly, double Lz)  
void SetPayload(TsPayloadV payload)
```

Argument

double M	Weight (kg)
double Lx	Offset of the center of gravity (mm)
double Ly	Offset of the center of gravity (mm)
double Lz	Offset of the center of gravity (mm)
double Gx	Work size(mm)
double Gy	Work size(mm)
double Gz	Work size(mm)
TsPayloadV payload	Specify the set value with TsPayLoadV

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

The load information that is set is inherited even if the master mode is changed.
To clear load data, specify SetPayload(0,0).

See also

SetTransTool()

TsRemoteV.method

Sets the tool coordinate.

Format

```
void SetTransTool(string name) *1
void SetTransTool(string name,double x,double y,double z,
                  double a, double b, double c) *2
void SetTransTool(string name,TsTransV tool) *2
```

Format

string name	*1 Specify the name of the coordinate type variable that is registered in the program file that is selected by the controller.
	*2 Names for the x,y,z, and c values that are set for convenience (displayed in TP)
double x	X set value
double y	Y set value
double z	Z set value
double a	A set value
double b	B set value
double c	C set value
TsTransV tool	Specify the set value with TsTrans.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

The tool coordinate that is set is inherited even if the master mode is changed.

See also

SetTransWork()

TsRemoteV.method

Sets the work coordinate.

Format

```
void SetTransWork(string name)*1
void SetTransWork(string name,double x,double y,double z,
                  double a, double b, double c)*2
void SetTransWork(string name,TsTransV work)*2
```

Argument

string name	*1 Specify the name of the coordinate type variable that is registered in the program file that is selected by the controller.
	*2 Names for the x,y,z, and c values that are set for convenience (displayed in TP)
double x	X set value
double y	Y set value
double z	Z set value
double a	A set value
double b	B set value
double c	C set value
TsTransV work	Specify the set value with TsTrans.

Return value

None

Exceptions

TCPNotConnect	Network is not connected
CommunicationError	Communication Error
ControllerNG	Command not accepted by controller
IllegalParam	Illegal parameter

Notice

The work coordinate that is set is inherited even if the master mode is changed.

See also

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